Operations Research

Midterm Review Guide

The five-station route through Lectures 1–5 on one example,
the 18 preparation exercises verified and annotated, and a mock midterm

Companion to the course of El Mehdi Saad

Every number in the exercise answers and in the mock midterm was recomputed exactly
(basis enumeration in rational arithmetic, `verify.py`).

How to use this guide

Part 1 walks the review deck's route, Model → Represent → Solve → Certify → Update, on the single production example, so that every method is seen once in full, with the statements she flags as traps collected in *Exam checks* boxes. **Part 2** does not repeat her 18 solutions; for each exercise it says what is being tested, the fastest safe route, the verified answer, the traps, and a twist she could add. **Part 3** is a two-hour mock midterm with one question per station and full solutions. The full textbook treatment is in `00_All_Chapters.pdf`; the proofs are in `OR_Theory_Review_and_Proof_Set.pdf`.

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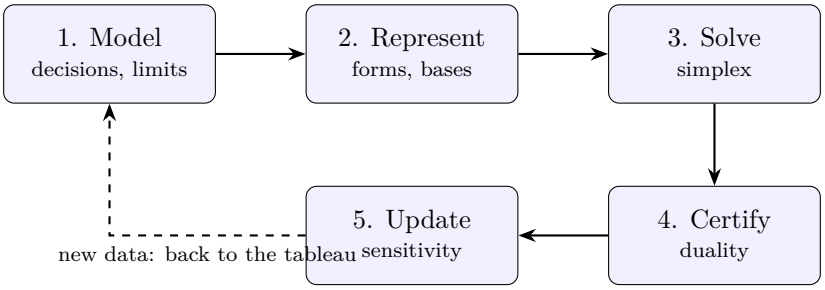
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Part I

The route: one example, five stations

1 The map



One production example connects the five stations. Each station answers one question: what is the decision (1), what is a candidate (2), how do we improve it (3), how do we *prove* we cannot improve further (4), and what survives when the data change (5). The deck's last slide says what a complete answer contains: *why the displayed equations justify the conclusion.* Every box below ends with that.

2 Station 1: Model

A mathematical program is $\max f(x)$ subject to $g_i(x) \le 0$, $x \in D$. Three things are defined, in this order: the **variables** (what we choose, each with a unit and a domain), the **objective** (what measures quality), the **constraints** (what is allowed). It is an **LP** when the objective and the constraints are linear and the variables continuous; integrality gives an IP, a product such as $x_1 x_2$ gives an NLP.

The running example

A workshop makes divisible products P and Q; all output sells. Per unit, P needs 2 h of resource 1 and 1 h of resource 2, Q needs 1 h and 2 h; 10 h and 8 h are available daily; net profits 500 and 400 MAD. With $x_1, x_2 \ge 0$ the daily amounts and profit in hundreds of MAD,

$$\max\ z = 5x_1 + 4x_2 \quad \text{s.t.} \quad 2x_1 + x_2 \le 10, \qquad x_1 + 2x_2 \le 8, \qquad x_1, x_2 \ge 0.$$

Dividing the objective by 100 changes the unit of z, not the optimizer; the resource rows keep their own units.

Technique: Linearising what does not look linear

Ratios: "Q must be at least 30 % of total output" is $x_2 \ge 0.3(x_1 + x_2)$, i.e. $0.3x_1 - 0.7x_2 \le 0$: linear once multiplied out. *Max and absolute value in the objective:* $\min \max_i |a + bt_i - y_i|$ becomes $\min \varepsilon$ with the $2n$ rows $\pm(a + bt_i - y_i) \le \varepsilon$ (exercise 15). *Fractions of projects:* a fraction $x_J \in [0, 1]$ is a continuous variable; its bounds are constraints with their own dual multipliers (exercise 8).

Exam checks: statements she can test

- Every variable is defined in words, with unit and domain, before the objective is written.
- "All output sells" is what justifies having no inventory variable; "divisible" is what justifies

continuity.

- $3x_1 + 2x_2$ is linear; x_1x_2, x_1^2, $\max(x_1, x_2)$ are not, but the last one is linearisable with an auxiliary variable.
- Scaling a row or the objective by a positive constant changes no optimizer, only the displayed numbers.

3 Station 2: Represent

3.1 Canonical and equational forms

Canonical (maximisation convention): $\max\{c^\top x : Ax \le b,\ x \ge 0\}$. Equational: $\max\{c^\top x : Ax = b,\ x \ge 0\}$. A row $a^\top x \le \beta$ becomes $a^\top x + s = \beta$ with a slack $s \ge 0$ (the unused amount); a row $a^\top x \ge \beta$ becomes $a^\top x - s = \beta$ with a surplus; an equality stays. The equational vector contains the original decisions *and* the slacks, in a stated order.

Original domain	Substitution	Nonnegative variables
$t \ge \ell$ (e.g. $-2 \le t \le 3$)	$u = t - \ell$	$t = u - 2,\ 0 \le u \le 5$
$t \le 0$	$t = -u$	$u \ge 0$
$t \in \mathbb{R}$ unrestricted	$t = u - v$	$u, v \ge 0$
minimise f	maximise $-f$	report $-$(optimal value)

Substitute everywhere the variable occurs and **track the constant** it creates in the objective (exercise 2: $\max 4u + 2v - 8$).

3.2 Matrix notation

Order the variables first: $x = (x_1, x_2, s_1, s_2)^\top$. Then

$$A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 10 \\ 8 \end{bmatrix}, \qquad c = \begin{bmatrix} 5 \\ 4 \\ 0 \\ 0 \end{bmatrix}, \qquad (Ax)_i = \sum_j a_{ij}x_j, \quad c^\top x = \sum_j c_j x_j.$$

Rows are equations, columns are variables. $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $x, c \in \mathbb{R}^n$.

3.3 Redundancy, consistency, rank

An inequality is **redundant** if the others imply it: $2x_1 + x_2 \le 10$ plus $x_1 + 2x_2 \le 8$ give $3x_1 + 3x_2 \le 18$, so adding that row changes nothing. The sufficient test is a nonnegative combination of the other rows that produces the candidate. To prove a row is *not* redundant, exhibit a point that satisfies all the others and violates it (exercise 4). Removing a redundant row before adding slacks saves a variable; keeping it is valid and gives the same optimal value.

For an equality system, eliminate on $[A \mid b]$: a row $0 = 0$ can be dropped; a row $0 = \beta \ne 0$ means the system is inconsistent; r pivots in the A block mean $\operatorname{rank} A = r$. When slack columns are present they contain I_m, so $\operatorname{rank} A = m$ at once.

Exam checks: statements she can test

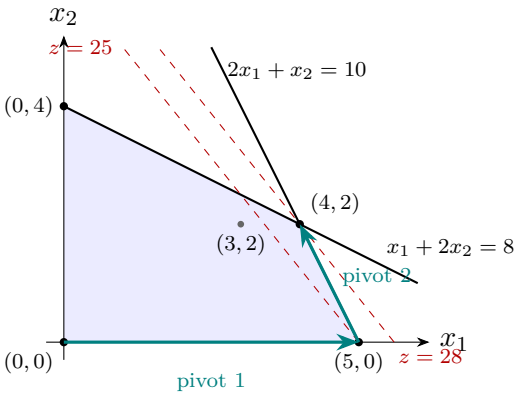
- The slack of a $\le$ row is added, the surplus of a $\ge$ row is subtracted; both are ≥ 0.

- A free variable costs two nonnegative ones; a variable with a lower bound costs a shift and a constant.
- "Redundant" is proved with a nonnegative combination; "not redundant" is proved with one violating point. Two proportional rows (exercise 9, mock Q1) are the easiest redundancy to miss.
- Dependent rows are removed *before* choosing a basis; a basis has exactly rank A columns.

4 Station 2b: Geometry and basic feasible solutions

4.1 Feasible, optimal, and the three outcomes

A **feasible** plan respects every constraint; an **optimal** plan has the best objective among feasible plans. $(3, 2)$ is feasible and earns 23; $(4, 2)$ earns 28.



A graphical answer identifies the region, its vertices, the optimal level line, and states the conclusion in words.

Outcome	Meaning	Example
Finite optimum	a best feasible value exists	the production LP
Unbounded objective	feasible values grow without limit	$\max x_1$, $x_1 \geq 0$
Infeasible	no plan satisfies all constraints	$x_1 \geq 2$, $x_1 \leq 1$

An unbounded feasible *set* can still have a finite optimum: $\max\{-x_1 : x_1 \geq 0\} = 0$. The question is always whether the objective *values* are bounded above.

4.2 Bases

Basis, basic solution, BFS

For $\max\{c^\top x : Ax = b, x \geq 0\}$ with $\operatorname{rank} A = m$, choose m independent columns in a stated order, $B = [A_{j_1} \cdots A_{j_m}]$. The **basic solution** is $x_N = 0$, $x_B = B^{-1}b$. It is a **BFS** exactly when $B^{-1}b \geq 0$, **nondegenerate** when all m basic coordinates are strictly positive. A BFS has *at most* m nonzero coordinates; basic coordinates may be zero. All of this concerns the full equational vector, slacks included.

For basic variables (x_1, x_2): $B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $B^{-1} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ (solve $Bq = v$ by hand: $q_1 =$

$(2v_1 - v_2)/3$, $q_2 = (-v_1 + 2v_2)/3)$, and $x_B = B^{-1}(10, 8)^\top = (4, 2)^\top \geq 0$. The full BFS is $(4, 2, 0, 0)$, objective 28: the zero coordinates here are the slacks, not the plotted ones.

Basic variables	x_1	x_2	s_1	s_2	Feasible?	z
x_1, x_2	4	2	0	0	yes	28
x_1, s_1	8	0	-6	0	no	–
x_1, s_2	5	0	0	3	yes	25
x_2, s_1	0	4	6	0	yes	16
x_2, s_2	0	10	0	-12	no	–
s_1, s_2	0	0	10	8	yes	0

The order of work in an enumeration: test column independence, solve, test nonnegativity, compare objectives *only* among the feasible ones.

Fundamental existence theorem

For an equational LP with independent rows and $x \geq 0$: if the feasible set is nonempty and the objective is bounded above, an optimal BFS exists. Hence enumeration of the at most $\binom{n}{m}$ column selections is a finite, correct method; several selections are dependent or infeasible and several bases may give the same (degenerate) point. Exercise 14 is the proof: a feasible point is a BFS iff the columns of its positive coordinates are independent; a dependent support can be shrunk along a null direction without decreasing $c^\top x$ (boundedness fixes the sign); the finitely many BFSs then contain a maximiser.

Exam checks: statements she can test

- "A BFS has exactly m nonzero coordinates" is false: at most m (degeneracy).
- An infeasible basic solution is still a basic solution; it is listed in the table and discarded only at the feasibility test.
- Two dependent columns (one a multiple of the other, exercise 3) are *not* a basis, even if the system happens to be solvable.
- A vertex where three lines meet in the plane (exercise 4, $A = (4, 0)$) is one point with several bases.

5 Station 3: Solve (simplex)

5.1 The three tableaux of the example

Start from the slack basis, which is feasible because $b \geq 0$:

$$\begin{aligned} s_1 &= 10 - 2x_1 - x_2 \\ s_2 &= 8 - x_1 - 2x_2 \\ z &= 0 + 5x_1 + 4x_2 \end{aligned}$$

Choose x_1 to enter (positive reduced cost; any positive one is allowed, the largest is the usual rule). With $x_2 = 0$: $s_1 \geq 0 \Rightarrow x_1 \leq 5$, $s_2 \geq 0 \Rightarrow x_1 \leq 8$; so s_1 leaves at $x_1 = 5$. Isolate $x_1 = 5 - \frac{1}{2}x_2 - \frac{1}{2}s_1$ and substitute in every row:

$$\begin{aligned} x_1 &= 5 - \tfrac{1}{2}x_2 - \tfrac{1}{2}s_1 \\ s_2 &= 3 - \tfrac{3}{2}x_2 + \tfrac{1}{2}s_1 \\ z &= 25 + \tfrac{3}{2}x_2 - \tfrac{5}{2}s_1 \end{aligned} \qquad \text{BFS } (5, 0, 0, 3),\ z = 25.$$

x_2 enters; $x_1 \geq 0 \Rightarrow x_2 \leq 10$, $s_2 \geq 0 \Rightarrow x_2 \leq 2$; s_2 leaves. Isolate $x_2 = 2 + \frac{1}{3}s_1 - \frac{2}{3}s_2$:

$$\begin{aligned} x_1 &= 4 - \tfrac{2}{3}s_1 + \tfrac{1}{3}s_2 \\ x_2 &= 2 + \tfrac{1}{3}s_1 - \tfrac{2}{3}s_2 \qquad (x_1, x_2) = (4, 2),\ z^\star = 28. \\ z &= 28 - 2s_1 - s_2 \end{aligned}$$

The last row is a **proof**: every feasible plan has $s_1, s_2 \geq 0$, hence $z = 28 - 2s_1 - s_2 \leq 28$, and the current BFS attains 28. In the plane the pivots walk $(0,0) \to (5,0) \to (4,2)$, each step keeping feasibility and increasing z.

5.2 The tableau in matrix form

Split $Ax = b$ into $Bx_B + A_N x_N = b$:

$$x_B = \underbrace{B^{-1}b}_{\bar{b}} - \underbrace{B^{-1}A_N}_{\bar{A}} x_N, \qquad z = \underbrace{c_B^\top B^{-1} b}_{z_0} + \sum_{j \in N} r_j x_j, \qquad r_j = c_j - c_B^\top B^{-1} A_j.$$

The displayed coefficient of x_j in a basic row is $-(B^{-1}A_j)$, because nonbasic terms sit on the right. Basic variables have reduced cost 0: $r_B^\top = c_B^\top - c_B^\top B^{-1} B = 0^\top$.

Basic direction and ratio test

Increase nonbasic x_j by $\theta \geq 0$, other nonbasics fixed at 0. The direction has $d_j = 1$, $d_i = 0$ for the other nonbasics, $d_B = -B^{-1}A_j$, and $Ad = 0$. Feasibility $x_B + \theta d_B \geq 0$ allows

$$\theta_{\max} = \min_{k:\, d_B(k) < 0} \frac{x_B(k)}{-d_B(k)}, \qquad \text{objective gain } \theta\, r_j.$$

At the first pivot's BFS, basis (x_1, s_2), entering x_2: $d_B = (-\frac{1}{2}, -\frac{3}{2})^\top$, $\theta_{\max} = \min\{5/\frac{1}{2},\ 3/\frac{3}{2}\} = 2$, gain $2 \cdot \frac{3}{2} = 3$: from 25 to 28.

The two conditions for an optimal basis (max convention)

$$\underbrace{B^{-1}b \geq 0}_{\text{feasibility: basic values}} \qquad \text{and} \qquad \underbrace{c^\top - c_B^\top B^{-1} A \leq 0^\top}_{\text{optimality: objective row}}.$$

Optimality theorem. If the basic solution is feasible and every nonbasic reduced cost is ≤ 0, it is optimal. Conversely, if a BFS is optimal *and nondegenerate*, its nonbasic reduced costs are ≤ 0. At a degenerate optimum a particular basis may fail the sign test (exercises 16d, 17d).

Unbounded improving direction	**Degenerate zero step**
$x = 1 + s$, $z = 1 + s$, $x, s \geq 0$: increasing s raises z and no row limits it. No limiting row: the objective is unbounded above.	$x + s = 0$, maximise $z = x$: $s = -x$, $z = x$. The basic value $s = 0$ blocks any increase of x: $\theta_{\max} = 0$. A positive reduced cost with no feasible positive step.

Reading uniqueness off the final row

If every nonbasic reduced cost is *strictly* negative, the optimum is unique: $z = z^\star$ forces all nonbasics to 0 and the rows then fix the basics (exercises 5b, 16b). If the optimal BFS is nondegenerate and some nonbasic reduced cost is 0, there are infinitely many optima: move along that basic direction for $0 \leq t \leq \theta_{\max}$ (exercises 12b, 16e, 18c). To list *all* optima, set the zero-cost nonbasic to t, keep the others at 0, and read the admissible interval of t from the rows.

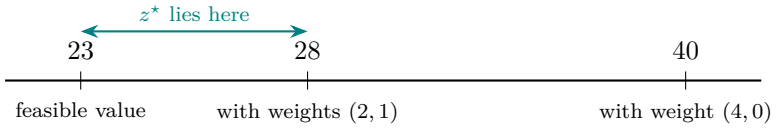
Exam checks: statements she can test

- The entering variable is a choice; the leaving variable is forced by the ratio test, taken only over rows whose coefficient is negative in the displayed form.
- "No limiting row" (unbounded) and "allowed step 0" (degenerate) are different; both can show a positive reduced cost.
- The optimality test is a property of the *basis*, not of the point: the same degenerate point can pass with one basis and fail with another (exercise 16d twist).
- A tableau with all reduced costs ≤ 0 but a negative basic value is *not* optimal: it is infeasible. Check both conditions, always.
- A zero reduced cost in a non-final tableau (mock Q3) means nothing yet: it is not an optimality statement.

6 Station 4: Certify (duality)

6.1 From a bound to the dual

The plan $\hat{x} = (3, 2)$ is feasible ($8 \leq 10$, $7 \leq 8$) and earns 23, so $23 \leq z^\star$. To know how much is left on the table we need an *upper* bound U: then $0 \leq z^\star - 23 \leq U - 23$.



By eye: $4 \times (2x_1 + x_2 \leq 10)$ gives $8x_1 + 4x_2 \leq 40$, and since $x_1 \geq 0$, $5x_1 + 4x_2 \leq 8x_1 + 4x_2 \leq 40$. Better: $2\times$ row $1 + 1\times$ row 2 gives exactly $5x_1 + 4x_2 \leq 28$. Weights $(1, 3)$ give 34. Which weights are valid, and which give the smallest bound? Call them $y_1, y_2 \geq 0$ (nonnegative so that multiplying keeps the direction):

$$(2y_1 + y_2)x_1 + (y_1 + 2y_2)x_2 \leq 10y_1 + 8y_2.$$

Because $x \geq 0$, the left side dominates the profit as soon as $2y_1 + y_2 \geq 5$ and $y_1 + 2y_2 \geq 4$ (coefficients may exceed 5 and 4; equality is not required). Minimising the bound under these conditions is the **dual**:

Primal: choose a plan	Dual: choose weights
$\max\ 5x_1 + 4x_2$	$\min\ 10y_1 + 8y_2$
$2x_1 + x_2 \leq 10,\ x_1 + 2x_2 \leq 8$	$2y_1 + y_2 \geq 5,\ y_1 + 2y_2 \geq 4$
$x \geq 0$	$y \geq 0$

$\max\ c^\top x$	$\min\ b^\top y$
$Ax \leq b$	$A^\top y \geq c$
$x \geq 0$	$y \geq 0$

One primal row gives one dual variable; one primal variable gives one dual row (sign restrictions excluded from the count). The transpose appears because the coefficient of x_j in the weighted sum is $\sum_i a_{ij} y_i = (A^\top y)_j$: read column j vertically.

Weak duality

For primal-feasible x and dual-feasible y: $c^\top x \leq (A^\top y)^\top x = y^\top A x \leq y^\top b$. The first step uses $x \geq 0$ and $A^\top y \geq c$; the second uses $y \geq 0$ and $Ax \leq b$. So $c^\top x \leq z^\star \leq b^\top y$.

A matching pair certifies optimality. $x = (4, 2)$: $10 \leq 10$, $8 \leq 8$, value 28. $y = (2, 1)$: $5 \geq 5$, $4 \geq 4$, value $20 + 8 = 28$. Weak duality gives $28 \leq z^\star \leq 28$: both are optimal, and the first plan $(3, 2)$ is now proved to lose exactly 5.

Strong duality

If the primal is feasible and bounded above, both problems have optimal solutions and $\max c^\top x = \min b^\top y$. Weak duality says every dual-feasible y is a valid bound; strong duality says an *exact* one exists. A matching pair proves optimality with weak duality alone.

Relaxation bounds. Dropping constraints enlarges the feasible set, so the relaxed optimum bounds the original: for a maximisation a feasible plan gives a lower bound and a relaxed optimum an upper bound (reverse for minimisation). Dropping resource 2 gives 40 at $(0, 10)$, a bound and *not* a feasible decision.

6.2 Mixed signs

Primal row	Dual variable	Primal variable	Dual row
$a_i^\top x \le b_i$	$y_i \ge 0$	$x_j \ge 0$	$A_j^\top y \ge c_j$
$a_i^\top x \ge b_i$	$y_i \le 0$	$x_j \le 0$	$A_j^\top y \le c_j$
$a_i^\top x = b_i$	$y_i \in \mathbb{R}$	$x_j \in \mathbb{R}$	$A_j^\top y = c_j$

(Max primal, min dual.) Row rule: choose the sign that turns the weighted row into a $\le$ inequality. Column rule: require $c_j x_j \le (A_j^\top y) x_j$ for the allowed signs of x_j. The dual objective is still $b^\top y$ with the original right-hand sides. Example: $\max 3x_1 + 2x_2$ s.t. $x_1 + x_2 \le 4$, $x_1 - x_2 = 1$, $x_1 \ge 0$, $x_2 \in \mathbb{R}$ has dual $\min 4y_1 + y_2$ s.t. $y_1 + y_2 \ge 3$, $y_1 - y_2 = 2$, $y_1 \ge 0$, $y_2 \in \mathbb{R}$.

6.3 Complementary slackness

Primal slack $s = b - Ax \ge 0$ (unused capacity); dual slack $t = A^\top y - c \ge 0$ (excess coefficient in the bound). Insert $y^\top Ax$ between the two objective values:

$$b^\top y - c^\top x = y^\top (b - Ax) + x^\top (A^\top y - c) = \sum_i y_i s_i + \sum_j x_j t_j, \qquad \text{every term } \ge 0.$$

Complementary slackness theorem

Feasible x and y are both optimal iff $y_i\,(b_i - (Ax)_i) = 0$ for all i and $x_j\,((A^\top y)_j - c_j) = 0$ for all j. Reading: $s_i > 0 \Rightarrow y_i = 0$; $y_i > 0 \Rightarrow s_i = 0$; $x_j > 0 \Rightarrow t_j = 0$; $t_j > 0 \Rightarrow x_j = 0$. At least one factor is zero; *both may be*: tightness alone does not force the paired variable to be positive. Feasibility on both sides is essential.

Technique: Finding the certificate without guessing

Take the feasible candidate $x = (4, 2)$. Both amounts are positive, so both dual rows must be tight: $2y_1 + y_2 = 5$, $y_1 + 2y_2 = 4$, giving $y = (2, 1)$. *Then verify*: $y \ge 0$, both dual rows hold, and $10(2) + 8(1) = 28 = 5(4) + 4(2)$. Complementary slackness finds candidates; feasibility plus matching values certifies them. If the system has no feasible solution, the candidate was not optimal (mock Q4 twist).

If we know that …	we may conclude …
one problem has a finite optimum	both do, with equal values
the primal is unbounded above	the dual is infeasible
the dual is unbounded below	the primal is infeasible
the primal is infeasible	the dual is unbounded *or* infeasible

Second row: a dual-feasible y would give a finite upper bound, contradicting unboundedness. Both can be infeasible: $\max x_1 + x_2$ s.t. $x_1 - x_2 \le -1$, $-x_1 + x_2 \le -1$, $x \ge 0$ (adding the rows: $0 \le -2$) has dual $\min -y_1 - y_2$ s.t. $y_1 - y_2 \ge 1$, $-y_1 + y_2 \ge 1$, $y \ge 0$ (adding: $0 \ge 2$).

Exam checks: statements she can test

- A feasible dual vector is a valid upper bound even if it is far from optimal; a feasible primal is a valid lower bound. "Gap > 0" does not prove the plan suboptimal.
- The dual variable of an equality row is free; the dual row of a free variable is an equality. The dual objective coefficient of y_i is the *original* b_i (even if the row was later rescaled).
- "Primal infeasible $\Rightarrow$ dual unbounded" is false (both infeasible is possible).
- A tight primal constraint can have $y_i = 0$ (exercise 8: the staff constraint is tight with multiplier 0).
- Weak duality is a two-line proof; be able to write it.

7 Station 5: Update (sensitivity analysis)

Start from the optimal tableau, basis (x_1, x_2), nonbasic (s_1, s_2):

$$x_1 = 4 - \tfrac{2}{3}s_1 + \tfrac{1}{3}s_2, \qquad x_2 = 2 + \tfrac{1}{3}s_1 - \tfrac{2}{3}s_2, \qquad z = 28 - 2s_1 - s_2.$$

Recover B^{-1} from the slack columns. The original slack columns form I_2, so the basic rows read $x_B = B^{-1}b - B^{-1}(s_1, s_2)^\top$: the displayed coefficients of (s_1, s_2) are $-B^{-1}$, hence

$$B^{-1} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \qquad c_B^\top B^{-1} = (5,4)B^{-1} = (2,1) = y^\top.$$

The multipliers are the negatives of the slack reduced costs: shadow prices 2 and 1 per hour of resources 1 and 2. Read rows in the stated basic order.

Changed data	What stays valid	What must be checked
new variable u (column a, profit γ)	old plan with $u = 0$	the new reduced cost $r_u = \gamma - c_B^\top B^{-1} a$
b changes	reduced costs	new basic values $B^{-1}b' \ge 0$
c changes	basic values	new reduced costs ≤ 0

These tests give the *range* in which the same basis stays optimal; outside it, reuse the tableau and pivot.

Case 1, new activity. $u \ge 0$, column $a = (1,1)^\top$, profit γ. $B^{-1}a = (\frac{1}{3}, \frac{1}{3})^\top$, $r_u = \gamma - (2,1)a = \gamma - 3$. Tableau: $x_1 = 4 - \frac{2}{3}s_1 + \frac{1}{3}s_2 - \frac{1}{3}u$, $x_2 = 2 + \frac{1}{3}s_1 - \frac{2}{3}s_2 - \frac{1}{3}u$, $z = 28 - 2s_1 - s_2 + (\gamma - 3)u$. For $\gamma = 2$ the old plan stays optimal with $u = 0$. For $\gamma = 4$, u enters; $x_2 = 2 - u/3$ hits zero first at $u = 6$; pivoting gives $x_1 = 2 - s_1 + s_2 + x_2$, $u = 6 + s_1 - 2s_2 - 3x_2$, $z = 34 - s_1 - 3s_2 - 3x_2$: new optimum $(x_1, x_2, u) = (2, 0, 6)$, $z^\star = 34$. A new variable adds a column, not an equation: B stays $m \times m$.

Case 2, resource change. $b' = (10 + \delta, 8)^\top$: $B^{-1}b' = (4 + \frac{2}{3}\delta,\ 2 - \frac{1}{3}\delta)^\top$. Reduced costs are unchanged (A, c unchanged), so the basis stays optimal iff both are ≥ 0: $-6 \le \delta \le 6$. There $z^\star(\delta) = 5(4 + \frac{2}{3}\delta) + 4(2 - \frac{1}{3}\delta) = 28 + 2\delta = 28 + y_1\delta$. For $\delta = 3$: plan $(6, 1)$, profit 34, *same basis, different plan.* For $\delta = 9$ the basic values are $(10, -1)$: infeasible. Reuse the tableau: the violated row $x_2 = -1 + \frac{1}{3}s_1 - \frac{2}{3}s_2$ is solved for $s_1 = 3 + 3x_2 + 2s_2$ and substituted: $x_1 = 8 - 2x_2 - s_2$, $s_1 = 3 + 3x_2 + 2s_2$, $z = 40 - 6x_2 - 5s_2$: optimum $(8, 0)$, $z^\star = 40$. The local formula $28 + 2\delta$ would have given 46: valid only inside its range.

Case 3, profit change. Profit of x_1 becomes κ. Feasible set and basic values $(4, 2)$ unchanged. Substitute the basic rows into the new objective:

$$z = \kappa(4 - \tfrac{2}{3}s_1 + \tfrac{1}{3}s_2) + 4(2 + \tfrac{1}{3}s_1 - \tfrac{2}{3}s_2) = 4\kappa + 8 + \tfrac{4-2\kappa}{3}s_1 + \tfrac{\kappa-8}{3}s_2,$$

optimal iff both coefficients are ≤ 0: $2 \leq \kappa \leq 8$. For $\kappa = 6$ the plan is still $(4, 2)$ but the profit is 32. At an endpoint another plan is also optimal. For $\kappa = 9$: $z = 44 - \frac{14}{3}s_1 + \frac{1}{3}s_2$; s_2 enters, $x_2 = 2 - \frac{2}{3}s_2$ leaves at $s_2 = 3$; new tableau $x_1 = 5 - \frac{1}{2}x_2 - \frac{1}{2}s_1$, $s_2 = 3 - \frac{3}{2}x_2 + \frac{1}{2}s_1$, $z = 45 - \frac{1}{2}x_2 - \frac{9}{2}s_1$: optimum $(5, 0)$, 45 (the old plan would earn 44).

Modification	Same optimal basis?	Same production?	Same profit?
new activity, $\gamma = 2$	yes	yes, $u = 0$	yes
capacity, $\delta = 3$	yes	no: $(6, 1)$	no: 34
profit, $\kappa = 6$	yes	yes: $(4, 2)$	no: 32

An optimal basis names *which* variables are basic; their values depend on b, their objective coefficients on c. Always state separately the modified plan and the modified value.

Simultaneous changes (exercise 17). With both b and c changed, the basis is optimal on the product of the two ranges, but $z^\star(\delta, \gamma) = c_B(\gamma)^\top B^{-1} b(\delta)$ contains a bilinear term: the total change is *not* the sum of the separate changes, because the resource change alters the quantity to which the new profit applies.

Exam checks: statements she can test

- A change in b never changes reduced costs; a change in c never changes basic values; a new column changes neither, it only adds a reduced cost.
- The shadow price y_i is the slope of $z^\star$ in b_i *inside the range only*; outside, pivot and the slope changes (it can drop to 0 when the resource stops binding, exercise 11).
- "The basis is unchanged" does not mean the plan or the profit is unchanged (table above).
- For a basic variable's profit there is often no upper limit on the range (mock Q5): raising its profit never makes it leave.
- The reduced cost of a slack is $-y_i$; a basic slack means $y_i = 0$.

8 Match the question to the method

What is requested	What the answer must contain
Formulate a model	variables with units and domains, objective, constraints, a sentence per constraint
Solve graphically	feasible region, labelled vertices, objective level line, the optimum in words
Enumerate BFSs	independence test, basic values, feasibility, objective values of the feasible ones
Use simplex	every tableau, the entering and leaving choice with its ratio test, the final row read as a certificate
Use duality	the dual written with the sign rules, feasibility of both candidates, the matching bound
Analyse a change	which of the two tests is affected, the valid range, the new plan *and* the new value
Prove a statement	the hypothesis used at each step (boundedness, nondegeneracy, feasibility of both sides)

The main ideas to keep together

1. Model first: what varies, what is allowed, what is optimised; units on everything.
2. Represent carefully: variable order, slack signs, constants from substitutions, rank before bases.
3. Solve with a justification: a feasible plan needs an optimality certificate, which the final row or a dual vector provides.
4. Read the tableau: basic values describe feasibility, reduced costs describe gains, slack reduced costs are $-y$.
5. Update what changed: keep the basis while its two conditions hold; beyond the range, pivot from the tableau.

Write the conclusion in the original decision variables and units.

Part II

The 18 preparation exercises, annotated

Her solutions are complete; what follows is the coach's reading of each one. *Tests* names the station and the skill. *Route* is the fastest safe path under exam time. *Answer* is the official result, recomputed exactly. *Traps* are the places where points are lost. *Twist* is the follow-up she could add, with its answer.

Exercise 1: a basic production model (Station 1)

Tests. Defining variables with units and domains; reading a resource table into rows.
Route. One variable per product, continuous because "divisible"; one row per resource, coefficients read down the product's column; objective from the profit row.
Answer (verified). $\max 50x_A + 40x_B$ s.t. $2x_A + x_B \le 8$, $x_A + 2x_B \le 7$, $x_A, x_B \ge 0$.
Traps. Writing "x_A = product A" instead of "units of A made per day". Forgetting to say why there is no inventory variable (every unit sells). Mixing the hour rows with the MAD objective.
Twist. "Solve it and certify." Vertices $(0,0), (4,0), (3,2), (0,3.5)$; values $0, 200, 230, 140$; optimum $(3,2)$, 230. Certificate: y from $2y_1 + y_2 = 50$, $y_1 + 2y_2 = 40$ gives $y = (20,10)$, $8(20) + 7(10) = 230$.

Exercise 2: an equivalent equational LP (Station 2)

Tests. Domain substitutions, slack versus surplus, the constant in the objective.
Route. Shift x by its lower bound, flip y, then add a slack to the $\le$ row and subtract a surplus from the $\ge$ row.
Answer (verified). $u = x + 2$, $v = -y$: $\max 4u + 2v - 8$ s.t. $2u + v + s_1 = 14$, $u - v - s_2 = -3$, $u, v, s_1, s_2 \ge 0$.
Traps. Dropping the -8 (the optimal *value* of the original is the new one minus 8). Adding the surplus instead of subtracting. Multiplying the second row by -1 to get a positive right-hand side is allowed but changes the sign of the surplus column: say which you did.
Twist. "Now y is free." $y = v_1 - v_2$ with $v_1, v_2 \ge 0$: the first row becomes $2u - v_1 + v_2 + s_1 = 14$, the objective $4u - 2v_1 + 2v_2 - 8$.

Exercise 3: enumerating basic solutions (Station 2b)

Tests. Rank from identity columns; independence before solving; feasibility before comparing; certifying with a row combination.
Route. Columns 1, 2 are I_2 so rank 2. For each of the $\binom{4}{2} = 6$ pairs: are the two columns proportional? If not, set the other two variables to 0 and solve the 2×2 system by substitution. Keep the feasible ones; compare.
Answer (verified). Pairs $(1,2), (1,3), (1,4)$ feasible with values $16, 14, 13$; $(2,3), (2,4)$ give $x_2 = -2$; $(3,4)$ dependent since $A_4 = 2A_3$. $z^\star = 16$ at $(4,2,0,0)$, certified by $3{\cdot}\text{row}_1 + 2{\cdot}\text{row}_2$: $c^\top x = 16 - x_3 - 3x_4 \le 16$.
Traps. Listing the dependent pair as "infeasible" (it is not a basis at all). Stopping at "the best of the three feasible is 16" without the certificate: enumeration proves optimality only through the fundamental theorem, and she asked for a proof.
Twist. "Is the optimum unique?" Yes: the certificate row has strictly negative coefficients on x_3, x_4, so $z = 16$ forces $x_3 = x_4 = 0$ and then $x_1 = 4, x_2 = 2$. "And if $c_4 = 10$?" Then $c^\top x = 16 - x_3$: every point with $x_3 = 0$, $x_1 = 4 - 2x_4$, $x_2 = 2 - 2x_4$, $0 \le x_4 \le 1$ is optimal: a segment of optima, value 16.

Exercise 4: graphical solution (Station 2b)

Tests. Redundancy proved both ways; vertices by intersecting lines; level lines.
Route. Test each row against the other two: a nonnegative combination proves redundancy, a violating point proves the opposite. Vertices: intersect pairs of boundary lines and keep those satisfying everything. Evaluate z at vertices.
Answer (verified). $x_1 \le 4$ is redundant ($3x_1 + x_2 \le 12$ and $x_2 \ge 0$). The other two are not ($(0,6)$ and $(4,3)$ are witnesses). Vertices $O(0,0)$, $A(4,0)$, $B(\frac{14}{5}, \frac{18}{5})$, $C(0,5)$; $z = 0, 12, \frac{78}{5}, 10$; unique optimum B, $z^\star = \frac{78}{5}$.
Traps. Claiming a row is redundant because it "looks weaker". Forgetting that $A = (4,0)$ lies on three lines: after slacks it is a degenerate BFS with several bases.
Twist. "Objective $3x_1 + 6x_2$." Parallel to $x_1 + 2x_2 = 10$: the whole edge BC is optimal, value 30; the optimum is not unique and the final tableau would show a zero reduced cost.

Exercise 5: simplex with three variables (Station 3)

Tests. Running the method cleanly; reading the full solution including slacks; uniqueness from strict signs.
Route. Largest reduced cost enters; ratio test over the rows where the entering variable has a negative displayed coefficient; rewrite every row. Two pivots.
Answer (verified). $(x_1, x_2, x_3, s_1, s_2, s_3) = (3, 4, 0, 0, 0, 5)$, $z^\star = 31$, unique because the final row $z = 31 - \frac{2}{5}x_3 - \frac{11}{5}s_1 - \frac{3}{5}s_2$ has strictly negative coefficients.
Traps. Forgetting to report $s_3 = 5$ (resource 3 is not binding). Ratio test with a wrong sign (a row where the entering variable has a positive coefficient never limits).
Twist. "Shadow prices?" $y = (\frac{11}{5}, \frac{3}{5}, 0)$, the negatives of the slack reduced costs; resource 3 has price 0 because $s_3 > 0$ (complementary slackness).

Exercise 6: solving through the dual (Station 4)

Tests. Writing a dual; spotting a redundant dual row; recovering the primal by complementary slackness; a uniqueness argument.
Route. Dual: one variable per resource, one row per product. The third dual row is implied by the second ($y_1 + 3y_2 \ge y_1 + 2y_2 \ge 5 > 4$). Solve the two-variable dual graphically; then: positive $y_i \Rightarrow$ primal row i tight; dual slack $t_3 > 0 \Rightarrow x_3 = 0$. Solve the 2×2 system.
Answer (verified). $y^\star = (\frac{7}{3}, \frac{4}{3})$, $w^\star = 54$; $x^\star = (4, 6, 0)$, value 54; unique because the complementary slackness conditions ($x_3 = 0$ and both rows tight) have a unique solution.
Traps. Saying the dual is unbounded because its region is unbounded (minimisation over an unbounded region can still have a finite optimum). Forgetting to *verify* the primal candidate's feasibility and value before concluding.
Twist. "Change c_3 to $\frac{17}{3}$." Then $t_3 = y_1 + 3y_2 - c_3 = 0$: x_3 may be positive and there are alternative optima; the dual optimum is unchanged, the primal optimal set becomes a segment.

Exercise 7: a tableau, a basis and a direction (Station 3 in matrix form)

Tests. Reading a given tableau; B and B^{-1} in the stated order; multipliers; a basic direction and its ratio test.
Route. Nonbasics to zero gives the point. B = the columns of the ordered basic variables in the equational A. $y^\top = c_B^\top B^{-1}$, or read $-$(slack reduced costs). The direction's basic part is $-B^{-1}A_3$, which is exactly the displayed column of x_3.
Answer (verified). $x = (4, 4, 0, 0, 0)$, feasible and optimal ($z = 28 - x_3 - s_1 - s_2$). $B = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} -1/5 & 2/5 \\ 3/5 & -1/5 \end{bmatrix}$, $y = (1, 1)$ with dual rows $4, 3, 3 \ge 4, 3, 2$ and value 28. $d =$

$(-\frac{3}{5}, -\frac{1}{5}, 1, 0, 0)$, $\theta_{\max} = \frac{20}{3}$, point $(0, \frac{8}{3}, \frac{20}{3}, 0, 0)$.
Traps. Wrong order of basic variables gives a transposed B. Reading y with the wrong sign (reduced cost of a slack is $-y_i$). Forgetting that $Ad = 0$ is *why* $d_B = -B^{-1}A_j$.
Twist. "Would simplex take this step?" No: $r_3 = -1 < 0$; the step is feasible but lowers z by $\frac{20}{3}$ to $\frac{64}{3}$. The direction exists for every nonbasic variable; the reduced cost decides whether it is worth taking.

Exercise 8: funding divisible projects (Stations 1 and 4)

Tests. Modelling coupling constraints and bounds; a hand-built dual certificate; uniqueness by tightness.
Route. Compare the benefit row with a multiple of the cost row; the multiple 2 matches or dominates every coefficient except C and E, which are fixed with their upper bounds $x_C \le 1$, $x_E \le 1$ at weight 1 each. Bound $= 2 \cdot 14 + 1 + 1 = 30$.
Answer (verified). Plan $(1, 0, 1, 1, 1)$ feasible, benefit 30, optimal; unique because equality forces the budget tight and $x_C = x_E = 1$, then $x_A = 1$, then $x_B = 0$, $x_D = 1$.
Traps. Forgetting that the bounds $x_J \le 1$ are constraints with their own multipliers. Using the staff row: it is tight ($7 = 7$) but its multiplier is 0; tightness does not force positivity. Giving the equality $x_B + x_D = 1$ a sign-restricted weight (its weight is free; here 0).
Twist. "Give the multipliers of all seven constraints and check complementary slackness." $y = (2, 0, 0, 0, 0)$ for budget, staff, $x_C \le x_A$, $x_B + x_D = 1$, $x_E \le x_B + x_C$; bounds: 1 on $x_C \le 1$ and $x_E \le 1$, 0 elsewhere. Every positive multiplier sits on a tight row; every positive x_J sits on a tight dual row (by construction the weighted sum equals the objective coefficient by coefficient).

Exercise 9: a redundant inequality and several bases (Stations 2 and 3)

Tests. Feasibility and redundancy as functions of a parameter, each proved in both directions; a simplex run where the "extra" row is the one that binds.
Route. Feasible iff the origin is: $\beta \ge 0$. Redundant iff $\beta \ge$ the maximum of $5x_1 + 5x_2$ under the other rows, which is 35 at $(3, 4)$ (the combination $2\cdot\text{row}_1 + \text{row}_2$ gives the bound, the point attains it). For $\beta = 30$ run simplex: s_3 leaves at the second pivot.
Answer (verified). Feasible iff $\beta \ge 0$; redundant iff $\beta \ge 35$. For $\beta = 30$: $(x_1, x_2, s_1, s_2, s_3) = (4, 2, 0, 5, 0)$, $z^\star = 28$.
Traps. Proving redundancy only for $\beta \ge 35$ and forgetting the converse (the point $(3, 4)$ is excluded when $\beta < 35$). Not noticing that at $\beta = 30$ the third row is tight and the second is slack: $y = (1, 0, \frac{3}{5})$, not $(2, 1, 0)$.
Twist. "For which β is the optimum 28?" The optimum of the first two rows, $(4, 2)$, has $5x_1 + 5x_2 = 30$; so $z^\star = 28$ iff $\beta \ge 30$, with different certificates below and above 35. For $\beta < 30$ the value drops (at $\beta = 25$: $x_1 + x_2 \le 5$ with $2x_1 + x_2 \le 10$ gives $(5, 0)$, $z = 25$).

Exercise 10: introducing a new activity (Station 5, case 1)

Tests. Transforming a new column by B^{-1}; the reduced cost as a function of p; continuing simplex; reading the dual from a 3×3 basis.
Route. B from the ordered basis (x_1, x_2, s_3); $B^{-1}a = (0, 1, 1)^\top$; $r_u = p - c_B^\top B^{-1}a = p - 5$. For $p = 10$: u enters, s_3 leaves at 4; then s_2 enters with reduced cost $\frac{1}{3}$, x_2 leaves at 2. Dual: $B^\top y = c_B$ for the final basis, or negatives of the nonbasic slack reduced costs with a basic slack giving 0.
Answer (verified). Old basis optimal iff $p \le 5$. For $p = 10$: $(x_1, x_2, x_3, u, s_1, s_2, s_3) = (\frac{20}{3}, 0, 0, \frac{14}{3}, 0, 2, 0)$, $z^\star = \frac{260}{3}$; dual $y = (\frac{2}{3}, 0, \frac{14}{3})$, dual rows $6, \frac{16}{3}, 10, 10$, value $\frac{260}{3}$.

Traps. Thinking the new activity changes m (it adds a column). Forgetting that u also enters the *constraint* rows through $B^{-1}a$. Stopping after the first pivot: a positive reduced cost ($\frac{1}{3}$ on s_2) remained.
Twist. "$p = 5$ exactly: all optima." $r_u = 0$; set $u = t$: $x_2 = 6 - t$, $s_3 = 4 - t$, so $0 \le t \le 4$ and the optimal set is $\{(6, 6-t, 0, t) : 0 \le t \le 4\}$, value 66.

Exercise 11: a resource change beyond the local interval (Station 5, case 2)

Tests. Reading one column of B^{-1}; intersecting three one-sided conditions; what to do outside the range; certificates in both regimes.
Route. First column of B^{-1} scales δ: basic values $6 - \frac{1}{5}\delta$, $7 + \frac{3}{5}\delta$, $2 - \frac{3}{5}\delta$. Nonnegativity gives $\delta \le 30$, $\delta \ge -\frac{35}{3}$, $\delta \le \frac{10}{3}$: intersect. Outside, find the new binding set by inspection (rows 2 and 3 tight) and rebuild the tableau.
Answer (verified). $-\frac{35}{3} \le \delta \le \frac{10}{3}$, $z^\star = 73 + \frac{17}{5}\delta$. $\delta = 2$: $(\frac{28}{5}, \frac{41}{5}, 0, 0, \frac{4}{5})$, $\frac{399}{5}$, $y = (\frac{17}{5}, \frac{1}{5}, 0)$. $\delta = 5$: old basis infeasible ($s_3 = -1$); new basis (x_1, x_2, s_1): $(\frac{16}{3}, 9, \frac{5}{3}, 0, 0)$, $\frac{253}{3}$, $y = (0, \frac{4}{3}, \frac{17}{3})$.
Traps. Taking only the binding condition from x_1 and missing that s_3 (the cap $x_2 \le 9$) is the one that limits. Using the local formula at $\delta = 5$ (it would give 90; the truth is $\frac{253}{3} \approx 84.3$). Not noticing the shadow price of resource 1 drops to 0 once it stops binding.
Twist. "Sketch $z^\star(\delta)$." Piecewise linear and concave: slope 4 for $\delta \le -\frac{35}{3}$ (only x_1, limited by resource 1), slope $\frac{17}{5}$ on the interval, slope 0 for $\delta \ge \frac{10}{3}$ (resource 1 idle). Continuity at the breakpoints: $\frac{100}{3}$ and $\frac{253}{3}$. Shadow prices of a resource never increase as the resource grows.

Exercise 12: changing a profit coefficient (Station 5, case 3)

Tests. Substituting a basic row into the new objective; a range from three conditions; complete optimal sets at endpoints; continuing simplex.
Route. $z_\gamma = z_0 + \gamma x_1$ with x_1 replaced by its row. Each nonbasic coefficient ≤ 0 gives one inequality in γ. At an endpoint set the zero-cost nonbasic to t and read the interval of t from the rows. For $\gamma = 2$, s_1 enters, x_2 leaves at $\frac{3}{2}$.
Answer (verified). $-\frac{3}{2} \le \gamma \le 1$; $z_\gamma = 39 + 6\gamma + (-3 - 2\gamma)x_3 + (-1 + \gamma)s_1 + (-2 - \gamma)s_2$. At $\gamma = -\frac{3}{2}$: $\{(6-2t, 3+t, t, 0, 0) : 0 \le t \le 3\}$, value 30. At $\gamma = 1$: $\{(6+t, 3-2t, 0, t, 0) : 0 \le t \le \frac{3}{2}\}$, value 45. $\gamma = 2$: $(\frac{15}{2}, 0, 0, \frac{3}{2}, 0)$, $z^\star = \frac{105}{2}$, $y = (0, \frac{7}{2})$.
Traps. Changing the constraint rows (only the objective row moves). Giving only the two endpoint BFSs instead of the whole segment. Forgetting the lower bound $\gamma \ge -2$ from s_2 is implied by $\gamma \ge -\frac{3}{2}$ but must still be checked.
Twist. "The profit of x_2 changes instead, to $3 + \gamma$." Substitute $x_2 = 3 + x_3 - 2s_1 + s_2$: $z = 39 + 3\gamma + (-3 + \gamma)x_3 + (-1 - 2\gamma)s_1 + (-2 + \gamma)s_2$, optimal iff $-\frac{1}{2} \le \gamma \le 2$.

Exercise 13: a parameter changes the status of an LP (Stations 2b and 4)

Tests. The three outcomes as functions of a parameter; the primal–dual status table; unbounded set versus unbounded objective versus unbounded optimal set.
Route. Rewrite the constraints as $0 \le x_2 \le \alpha$, $x_1 \ge x_2 + 1$. Infeasible iff $\alpha < 0$. For the sign of x_1's coefficient: negative means x_1 as small as allowed ($x_1 = x_2 + 1$), zero means x_1 free to grow with no gain, positive means unbounded. Dual by the rules; check each regime against the table.
Answer (verified). $\alpha < 0$ infeasible; $0 \le \alpha \le 1$ finite, $z^\star = \alpha^2 + \alpha - 1$ at $(\alpha + 1, \alpha)$, and at $\alpha = 1$ the whole ray $\{(x_1, 1) : x_1 \ge 2\}$; $\alpha > 1$ unbounded. Dual: infeasible for $\alpha > 1$, unbounded below for $\alpha < 0$, finite with $y = (1 - \alpha, \alpha)$ otherwise. The feasible set is unbounded for every $\alpha \ge 0$; the optimal set is unbounded only at $\alpha = 1$.

Traps. Confusing the three "unbounded"s. Forgetting that at $\alpha = 1$ the optimum is attained (value 1) although the optimal set is a ray.
Twist. "Why is the dual infeasible exactly when the primal is unbounded here, when the table only says 'may'?" Because the primal is feasible for $\alpha > 1$: a feasible primal that is unbounded forces dual infeasibility; the "may" in the table covers the case where the primal is infeasible.

Exercise 14: why an optimal BFS exists (Station 2b, proof)

Tests. The one proof she can ask in full: support independence characterises BFSs; shrinking a support; finiteness.
Route. (a) BFS $\Rightarrow$ support $\subseteq$ basis $\Rightarrow$ independent; independent support $\Rightarrow$ extend to m columns using $\operatorname{rank} A = m$, then $x_I = A_I^{-1} b$. (b) Dependent support gives $d \neq 0$ with $Ad = 0$ supported on S; choose the sign so $c^\top d \geq 0$; boundedness forbids $d \geq 0$ with $c^\top d > 0$, so some $d_j < 0$; step $t^\star = \min_{d_j<0} x_j/(-d_j) > 0$ kills a coordinate, keeps feasibility, does not decrease $c^\top x$. (c) Finitely many BFSs; the best one beats every feasible point through (b).
Traps. Using boundedness nowhere (then the proof is wrong: $\max x_1$ on $x_1 \geq 0$ has no optimal BFS). Forgetting to argue $t^\star > 0$ (numerators are positive because d lives on the support). Claiming "at most $\binom{n}{m}$" without saying each basis gives at most one point.
Twist. "Where does it fail if $\operatorname{rank} A < m$?" In (a), the extension to m independent columns is impossible; the fix is to delete dependent rows first, which does not change the feasible set.

Exercise 15: uniform approximation by a line (Stations 1 and 4)

Tests. Linearising a max of absolute values; a dual of a minimisation with free variables; a certificate by cancellation.
Route. $\min \varepsilon$ with six rows $\pm(a + bt_i - y_i) \leq \varepsilon$. Lower bound: combine three rows with weights that cancel both a and b: $(1, 2, 1)$ on the rows $a - 1$, $4 - a - b$, $a + 2b - 2$ gives $5 \leq 4\varepsilon$. Attain it by making those three rows tight. Dual by the rules: free a, b give two equalities; $\varepsilon \geq 0$ with cost 1 gives $\sum(u_i + v_i) \leq 1$.
Answer (verified). $a = \frac{9}{4}$, $b = \frac{1}{2}$, $E_{\min} = \frac{5}{4}$; residuals $(\frac{5}{4}, -\frac{5}{4}, \frac{5}{4})$. Dual certificate $u = (\frac{1}{4}, 0, \frac{1}{4})$, $v = (0, \frac{1}{2}, 0)$, value $\frac{5}{4}$. Unique line.
Traps. Keeping max or $|\cdot|$ in the model. Writing the dual of a *maximisation* by reflex: here the primal is a min, the dual a max, and the row signs flip accordingly. Not checking the three other rows after fixing (a, b).
Twist. "Why do the residuals alternate in sign?" Because the certificate weights $(1, 2, 1)$ multiply rows of alternating orientation $(+, -, +)$; this equioscillation is the signature of a best uniform approximation. "Least absolute deviations instead?" $\min \sum_i (e_i^+ + e_i^-)$ s.t. $a + bt_i - y_i = e_i^+ - e_i^-$, $e^\pm \geq 0$.

Exercise 16: reduced costs, what can and cannot be concluded (Station 3, theory)

Tests. The identity $c^\top w = c_B^\top B^{-1} b + \sum_{j \in N} \bar{c}_j w_j$ and everything that follows from it; a counterexample.
Route. (a) $\bar{c} \leq 0$ and $w \geq 0$ give $c^\top w \leq c^\top x$. (b) Strict signs force $w_N = 0$, then $w_B = x_B$. (c) A positive $\bar{c}_j$ plus nondegeneracy gives a feasible improving step: contradiction. (d) $\max x_2$ s.t. $x_1 + x_2 = 0$, basis $\{x_1\}$: point $(0, 0)$ optimal, degenerate, $\bar{c}_2 = 1 > 0$. (e) Zero reduced cost plus nondegeneracy gives a feasible segment of equal value.
Traps. Proving (c) without using nondegeneracy (then (d) contradicts you). In (d), claiming the point is not optimal: it is the only feasible point. Confusing "basis fails the test" with "point is not optimal".
Twist. "In (d), take x_2 as the basic variable." $B = (1)$, $x_B = 0$, $\bar{c}_1 = 0 - 1 \cdot 1 = -1 \leq 0$: the test passes. Same point, other basis: the test is about bases.

Exercise 17: simultaneous changes in a resource and a profit (Station 5)

Tests. Both tests at once; the bilinear value function; a degenerate point where the basis test fails but the point is optimal.
Route. $x_B = B^{-1}b(\delta)$ gives the feasibility range in δ; $y^\top = c_B(\gamma)^\top B^{-1}$ gives slack reduced costs $-y$ and the range in γ. Multiply out $z^\star = c_B(\gamma)^\top x_B(\delta)$. For $(20,-3)$: compute the basic solution, see $x_1 = 0$, then certify $(0,16)$ by a direct bound.
Answer (verified). $x = (4-\frac{1}{5}\delta, 4+\frac{3}{5}\delta, 0, 0)$, $\bar{c}_{s_1} = -1+\frac{1}{5}\gamma$, $\bar{c}_{s_2} = -1-\frac{2}{5}\gamma$; region $[-\frac{20}{3}, 20] \times [-\frac{5}{2}, 5]$; $z^\star = 28 + 4\gamma + \delta - \frac{1}{5}\gamma\delta$. Values $28, 33, 36, 39$: simultaneous change 11, sum of separate changes 13, difference $-\frac{1}{5}\cdot 2\cdot 5 = -2$. At $(20,-3)$ the basis fails ($\bar{c}_{s_2} = \frac{1}{5}$) but $(0,16)$ is optimal: $x_1 + 3x_2 \le 3(3x_1 + x_2) \le 48$, attained; unique.
Traps. Expecting additivity. Concluding "not optimal" from the failed test at a degenerate point. Forgetting that the region is a rectangle only because δ and γ act on different tests.
Twist. "Which basis passes the test at $(20,-3)$?" Basis (x_2, s_1): $B = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$, $y = (3,0)B^{-1} = (0,3)$, reduced costs x_1: $1 - 9 = -8$, s_2: -3; both ≤ 0. The degenerate vertex $(0,16)$ has three bases; one of them certifies it.

Exercise 18: a new activity with an unknown profit (Station 5, parametric)

Tests. A full parametric analysis: thresholds, equality cases, value function, a dual certificate per regime.
Route. Solve the two-variable LP (two pivots). $B^{-1}a = (\frac{2}{5}, \frac{1}{5})^\top$, $r_u = p - \frac{13}{5}$. Above the threshold, pivot u in; the next tableau's reduced costs are affine in p and give the next threshold, 5. Duals: read the final row of each regime.
Answer (verified). $z^\star(p) = 44$ for $0 \le p \le \frac{13}{5}$ at $(6,4,0)$; $5 + 15p$ for $\frac{13}{5} \le p \le 5$ at $(0,1,15)$; $16p$ for $p \ge 5$ at $(0,0,16)$. At $p = \frac{13}{5}$: $\{(6 - \frac{2}{5}t, 4 - \frac{1}{5}t, t) : 0 \le t \le 15\}$; at $p = 5$: $\{(0, t, 16 - t) : 0 \le t \le 1\}$. Duals $(\frac{7}{5}, \frac{6}{5})$, $(\frac{3p-5}{2}, \frac{5-p}{2})$, $(p, 0)$.
Traps. Reporting only BFSs at the thresholds. Not checking continuity of $z^\star$ at $\frac{13}{5}$ and 5 (both formulas agree). Forgetting that the second tableau's reduced costs still depend on p.
Twist. "Why is $z^\star(p)$ convex?" It is the maximum over the (fixed) feasible set of the affine functions $p \mapsto 4x_1 + 5x_2 + pu$: a maximum of affine functions is convex. Its slopes $0, 15, 16$ are the optimal u, nondecreasing in p.

What the set as a whole says about the exam

Each station is tested at least twice, and the hard questions are always the second half of an exercise: *prove* redundancy both ways (4, 9), *certify* with a hand-built combination (3, 8, 15, 17d), *list all* optima at a threshold (10, 12, 18), *go outside* the range and pivot (10, 11, 12), and *separate* what is true of a basis from what is true of a point (16, 17). Exercise 14 is the one proof likely to be asked in full.

Part III

Mock midterm

9 The paper (2 hours, 20 points)

One question per station. Work in order; the tableau of Q3 is reused in Q5.

Q1. Model and represent (3 pts)

A bakery sells baguettes (x_1 per day) and whole-grain loaves (x_2 per day), both divisible for planning purposes. A baguette needs 0.25 kg of flour and 1 minute of oven time; a loaf needs 0.5 kg and 2 minutes. 120 kg of flour and 400 oven-minutes are available daily. At least 25 % of the units baked must be loaves. Profits are 2 and 3 MAD per unit.

(a) Formulate an LP. State the unit of every row.

(b) One resource row is redundant. Identify it and prove it.

(c) Write the equational form after removing the redundant row, state the variable order, and give A, b, c.

(d) Solve graphically.

Q2. Bases and certificates (3 pts)

Let $A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$, $c = (3, 1, 0, 0)^\top$, and consider $\max\{c^\top x : Ax = b,\ x \geq 0\}$.

(a) Give $\operatorname{rank} A$ with a justification.

(b) For each of the six column pairs, state whether it is a basis; if so, give the full basic solution and its feasibility, in a table.

(c) Give the optimal value and prove it with a combination of the two rows.

(d) Is the optimum unique? Prove your answer.

Q3. Simplex (4 pts)

$$\max\ 3x_1{+}2x_2{+}4x_3 \quad \text{s.t.} \quad x_1{+}x_2{+}2x_3 \leq 4, \qquad 2x_1{+}3x_3 \leq 5, \qquad 2x_1{+}x_2{+}3x_3 \leq 7, \qquad x \geq 0.$$

(a) Solve by simplex from the slack basis, entering the variable with the largest reduced cost. Show every tableau and every ratio test.

(b) One intermediate tableau has a nonbasic reduced cost equal to zero. Say what this does and does not mean at that stage.

(c) Give the full optimal solution (decisions and slacks), the optimal value, and decide whether the optimum is unique.

(d) Read the dual optimal solution off the final tableau and verify it in the dual.

Q4. Duality with mixed signs (4 pts)

$$\max\ 2x_1 + x_2 - x_3$$
$$\text{s.t. } x_1 - x_2 + x_3 \le 8, \qquad x_1 - x_2 \ge 2, \qquad x_1 + 2x_3 = 4, \qquad x_1 \ge 0,\ x_2 \le 0,\ x_3 \in \mathbb{R}.$$

(a) Write the dual, justifying the sign of each dual variable and the direction of each dual row.

(b) Is $\hat{x} = (12, 0, -4)$ feasible? Use complementary slackness to construct the only possible dual certificate, verify it, and conclude.

(c) Now replace the first constraint by $x_1 + x_2 + x_3 \le 8$. Exhibit a feasible ray along which the objective is unbounded. What does this imply for the new dual? Confirm it directly from the new dual constraints.

Q5. Sensitivity on the tableau of Q3 (4 pts)

Use the final tableau of Q3, with the ordered basis you obtained.

(a) Give B^{-1} and $y^\top = c_B^\top B^{-1}$ by reading the slack columns.

(b) The second right-hand side becomes $5 + \delta$. For which δ does the basis stay optimal, and what is $z^\star(\delta)$ there?

(c) For $\delta = 2$, show the old basis is infeasible, reoptimise from the tableau, and give the new optimal solution, value and dual certificate.

(d) A new product $u \ge 0$ uses one unit of each resource and earns p. For which p does the old basis stay optimal? For $p = 3$, find the new optimum.

(e) The profit of x_1 becomes $3 + \gamma$. Give the range of γ for which the basis stays optimal and explain why it has no upper end.

Q6. True or false, with a one-line reason each (2 pts)

1. An LP whose feasible region is unbounded has no finite optimum.
2. A basic feasible solution has exactly m nonzero coordinates.
3. If a basis is feasible and all its nonbasic reduced costs are ≤ 0, its basic solution is optimal.
4. If a basic feasible solution is optimal, every basis representing it has nonbasic reduced costs ≤ 0.
5. If the primal is infeasible, the dual is unbounded.
6. At an optimal pair, a tight primal constraint has a positive dual variable.
7. Changing b leaves the reduced costs of the current basis unchanged.
8. The optimal-value change caused by two simultaneous data changes is the sum of the two separate changes.
9. A nondegenerate optimal BFS with a zero nonbasic reduced cost implies infinitely many optimal solutions.
10. Removing a redundant constraint can change the optimal value.

10 Solutions

Q1

(a) $x_1, x_2 \geq 0$ units per day.

$$\begin{aligned} &\max\ 2x_1 + 3x_2 \text{ [MAD]} \\ &\text{s.t. } 0.25x_1 + 0.5x_2 \leq 120 \text{ [kg]}, \qquad x_1 + 2x_2 \leq 400 \text{ [min]}, \\ &\qquad x_2 \geq 0.25(x_1 + x_2) \ \Leftrightarrow\ x_1 - 3x_2 \leq 0 \text{ [units]}. \end{aligned}$$

(b) The flour row times 4 is $x_1 + 2x_2 \leq 480$, which the oven row implies ($x_1 + 2x_2 \leq 400 \leq 480$). The flour row is redundant: a nonnegative combination (weight $\frac{1}{4}$ of the oven row, plus nothing) yields it. The oven row is not redundant: $(400, 0)\cdot$ scaled, e.g. $(480, 0)$, satisfies flour and the ratio row but not the oven row.

(c) Order (x_1, x_2, s_1, s_2): $x_1 + 2x_2 + s_1 = 400$, $x_1 - 3x_2 + s_2 = 0$,

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 1 & -3 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 400 \\ 0 \end{bmatrix}, \quad c = \begin{bmatrix} 2 \\ 3 \\ 0 \\ 0 \end{bmatrix}.$$

(d) Region: below $x_1 + 2x_2 = 400$, above $x_1 = 3x_2$ (i.e. $x_2 \geq x_1/3$), first quadrant. Vertices $(0, 0)$, $(0, 200)$, and the intersection $3x_2 + 2x_2 = 400 \Rightarrow (240, 80)$. Values 0, 600, 720. Optimum: bake 240 baguettes and 80 loaves for 720 MAD; the level line $2x_1 + 3x_2 = 720$ touches the region only at $(240, 80)$, so the optimum is unique. (Check: oven $240 + 160 = 400$ tight; ratio $80 = 0.25 \cdot 320$ tight; the right-hand side 0 of the ratio row makes the origin a degenerate vertex.)

Q2

(a) Columns 3 and 4 are e_1, e_2: independent, so $\operatorname{rank} A = 2$ (two rows).

(b) No two columns are proportional, so all six pairs are bases.

Columns	Solve	Full basic solution	z	Feasible?
1,2	$x_1 + x_2 = 6$, $x_1 - x_2 = 2$	$(4, 2, 0, 0)$	14	yes
1,3	$x_1 = 2$, $x_3 = 4$	$(2, 0, 4, 0)$	6	yes
1,4	$x_1 = 6$, $x_4 = -4$	$(6, 0, 0, -4)$	–	no
2,3	$x_2 = -2$, $x_3 = 8$	$(0, -2, 8, 0)$	–	no
2,4	$x_2 = 6$, $x_4 = 8$	$(0, 6, 0, 8)$	6	yes
3,4	$x_3 = 6$, $x_4 = 2$	$(0, 0, 6, 2)$	0	yes

(c) $2\cdot\text{row}_1 + 1\cdot\text{row}_2$: $3x_1 + x_2 + 2x_3 + x_4 = 14$, so $c^\top x = 14 - 2x_3 - x_4 \leq 14$ for every feasible x; $(4, 2, 0, 0)$ attains it. $z^\star = 14$.

(d) Unique: equality in (c) forces $x_3 = x_4 = 0$, and then the two rows give $x_1 = 4$, $x_2 = 2$.

Q3

(a) Tableau 0, basis (s_1, s_2, s_3):

$$\begin{aligned} s_1 &= 4 - x_1 - x_2 - 2x_3, \quad & s_2 &= 5 - 2x_1 - 3x_3, \\ s_3 &= 7 - 2x_1 - x_2 - 3x_3, \quad & z &= 0 + 3x_1 + 2x_2 + 4x_3. \end{aligned}$$

x_3 enters (largest, 4). Ratios: $4/2 = 2$, $5/3$, $7/3$; s_2 leaves at $\frac{5}{3}$. Tableau 1, basis (s_1, x_3, s_3):

$$s_1 = \tfrac{2}{3} + \tfrac{1}{3}x_1 - x_2 + \tfrac{2}{3}s_2, \quad x_3 = \tfrac{5}{3} - \tfrac{2}{3}x_1 - \tfrac{1}{3}s_2,$$
$$s_3 = 2 - x_2 + s_2, \qquad\qquad z = \tfrac{20}{3} + \tfrac{1}{3}x_1 + 2x_2 - \tfrac{4}{3}s_2.$$

x_2 enters (2). Ratios: s_1: $\frac{2}{3}$, s_3: 2 (x_3 row has no x_2); s_1 leaves at $\frac{2}{3}$. Tableau 2, basis (x_2, x_3, s_3):

$$x_2 = \tfrac{2}{3} + \tfrac{1}{3}x_1 - s_1 + \tfrac{2}{3}s_2, \quad x_3 = \tfrac{5}{3} - \tfrac{2}{3}x_1 - \tfrac{1}{3}s_2,$$
$$s_3 = \tfrac{4}{3} - \tfrac{1}{3}x_1 + s_1 + \tfrac{1}{3}s_2, \quad z = 8 + x_1 - 2s_1 + 0 \cdot s_2.$$

x_1 enters (1). Ratios: x_3: $\frac{5}{3}/\frac{2}{3} = \frac{5}{2}$, s_3: $\frac{4}{3}/\frac{1}{3} = 4$ (x_2 row increases); x_3 leaves at $\frac{5}{2}$. Tableau 3, basis (x_2, x_1, s_3):

$$x_2 = \tfrac{3}{2} - \tfrac{1}{2}x_3 - s_1 + \tfrac{1}{2}s_2, \quad x_1 = \tfrac{5}{2} - \tfrac{3}{2}x_3 - \tfrac{1}{2}s_2,$$
$$s_3 = \tfrac{1}{2} + \tfrac{1}{2}x_3 + s_1 + \tfrac{1}{2}s_2, \quad z = \tfrac{21}{2} - \tfrac{3}{2}x_3 - 2s_1 - \tfrac{1}{2}s_2.$$

All reduced costs ≤ 0, all basic values ≥ 0: optimal.
(b) In tableau 2 the reduced cost of s_2 is 0 while x_1 has reduced cost $1 > 0$. The tableau is not optimal, so the zero says nothing about alternative optima; it only says that moving along the s_2 direction from *this* BFS would not change z. Zero reduced costs matter only in an optimal tableau (and then only at a nondegenerate BFS).
(c) $(x_1, x_2, x_3, s_1, s_2, s_3) = (\frac{5}{2}, \frac{3}{2}, 0, 0, 0, \frac{1}{2})$, $z^\star = \frac{21}{2}$. Unique: the three nonbasic reduced costs are strictly negative, so $z = \frac{21}{2}$ forces $x_3 = s_1 = s_2 = 0$ and the rows fix the rest.
(d) Slack reduced costs $(-2, -\frac{1}{2}, 0)$ give $y = (2, \frac{1}{2}, 0)$ (s_3 basic, so $y_3 = 0$). Dual: $\min 4y_1 + 5y_2 + 7y_3$ s.t. $y_1 + 2y_2 + 2y_3 \ge 3$, $y_1 + y_3 \ge 2$, $2y_1 + 3y_2 + 3y_3 \ge 4$, $y \ge 0$. Check: $2 + 1 = 3$, $2 \ge 2$, $4 + \frac{3}{2} = \frac{11}{2} \ge 4$; value $8 + \frac{5}{2} = \frac{21}{2}$. Matches.

Q4

(a) Row 1 is $\le$: $y_1 \ge 0$ (multiplying by $y_1 \ge 0$ keeps $\le$). Row 2 is $\ge$: $y_2 \le 0$ (a nonpositive weight turns it into $\le$). Row 3 is $=$: $y_3 \in \mathbb{R}$. Column of $x_1 \ge 0$, $(1, 1, 1)$: need $y_1 + y_2 + y_3 \ge 2$. Column of $x_2 \le 0$, $(-1, -1, 0)$: need $(A_2^\top y)x_2 \ge c_2 x_2$ for $x_2 \le 0$, i.e. $A_2^\top y \le c_2$: $-y_1 - y_2 \le 1$. Column of x_3 free, $(1, 0, 2)$: equality $y_1 + 2y_3 = -1$. Dual:

$$\min\ 8y_1 + 2y_2 + 4y_3$$
$$\text{s.t. } y_1 + y_2 + y_3 \ge 2, \qquad -y_1 - y_2 \le 1, \qquad y_1 + 2y_3 = -1, \qquad y_1 \ge 0,\ y_2 \le 0,\ y_3 \in \mathbb{R}.$$

(b) $\hat{x} = (12, 0, -4)$: row 1: $12 - 0 - 4 = 8 \le 8$ tight; row 2: $12 \ge 2$ slack; row 3: $12 - 8 = 4$; signs fine. Feasible, value $24 + 0 + 4 = 28$. Complementary slackness: row 2 slack $\Rightarrow y_2 = 0$; $x_1 = 12 > 0 \Rightarrow$ its dual row tight: $y_1 + y_3 = 2$; x_3 free $\Rightarrow$ its row is an equality anyway: $y_1 + 2y_3 = -1$. Hence $y_3 = -3$, $y_1 = 5$, $y = (5, 0, -3)$. Verify: $y_1 \ge 0$, $y_2 \le 0$; row 2 of the dual: $-5 \le 1$; value $40 + 0 - 12 = 28$. Primal and dual feasible with equal values: both optimal, $z^\star = 28$.
(c) From $\hat{x}$ take $d = (2, -1, -1)$: row 1 changes by $2 - 1 - 1 = 0$, row 2 by $2 + 1 = 3 > 0$ (allowed, it is a $\ge$ row), row 3 by $2 - 2 = 0$; x_1 increases, x_2 decreases (allowed, $x_2 \le 0$), x_3 is free. So $\hat{x} + td$ is feasible for all $t \ge 0$ and the objective changes by $t(4 - 1 + 1) = 4t \to \infty$: the primal is unbounded above, hence the dual must be infeasible. Directly: the new dual row for x_2 (column $(1, -1, 0)$, $x_2 \le 0$) is $y_1 - y_2 \le 1$. With $y_1 + 2y_3 = -1$ and $y_1 + y_2 + y_3 \ge 2$: $y_3 = -(1 + y_1)/2$, so $\frac{1}{2}y_1 + y_2 \ge \frac{5}{2}$; with $y_2 \le 0$ this needs $y_1 \ge 5$, and then $y_1 - y_2 \ge 5 > 1$. No dual-feasible point exists.

Q5

(a) Ordered basis (x_2, x_1, s_3). The slack coefficients in the three rows are $-B^{-1}$:

$$B^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 \\ -1 & -\frac{1}{2} & 1 \end{bmatrix}, \qquad y^\top = (2,3,0)B^{-1} = (2, \tfrac{1}{2}, 0),$$

also the negatives of the slack reduced costs.
(b) $b' = b + \delta e_2$: basic values change by $\delta\cdot$(second column of B^{-1}) $= (-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})\delta$: $x_2 = \frac{3}{2} - \frac{1}{2}\delta$, $x_1 = \frac{5}{2} + \frac{1}{2}\delta$, $s_3 = \frac{1}{2} - \frac{1}{2}\delta$. Nonnegativity: $\delta \le 3$, $\delta \ge -5$, $\delta \le 1$; range $-5 \le \delta \le 1$. Reduced costs unchanged, so there $z^\star(\delta) = \frac{21}{2} + y_2\delta = \frac{21}{2} + \frac{1}{2}\delta$.
(c) $\delta = 2$: $(x_2, x_1, s_3) = (\frac{1}{2}, \frac{7}{2}, -\frac{1}{2})$, infeasible. The violated row reads $s_3 = -\frac{1}{2} + \frac{1}{2}x_3 + s_1 + \frac{1}{2}s_2$; to make it feasible, bring in a nonbasic with a positive coefficient there, choosing the one that keeps the objective row nonpositive: ratios $|r_j|$/coef are x_3: 3, s_1: 2, s_2: 1; s_2 enters, s_3 leaves: $s_2 = 1 - x_3 - 2s_1 + 2s_3$. Substituting (with the $\delta = 2$ constants $x_2 = \frac{1}{2} - \ldots$, $x_1 = \frac{7}{2} - \ldots$):

$$x_2 = 1 - x_3 - 2s_1 + s_3, \quad x_1 = 3 - x_3 + s_1 - s_3,$$
$$s_2 = 1 - x_3 - 2s_1 + 2s_3, \quad z = 11 - x_3 - s_1 - s_3.$$

Optimal: $(x_1, x_2, x_3) = (3, 1, 0)$, $s_2 = 1$, $z^\star = 11$. Dual: $y = (1, 0, 1)$ (nonbasic slacks s_1, s_3 with reduced costs $-1, -1$; s_2 basic). Check: $1 + 0 + 2 = 3 \ge 3$, $1 + 1 = 2 \ge 2$, $2 + 0 + 3 = 5 \ge 4$; value $4 + 0 + 7 = 11$. Note that the local formula would have given $\frac{23}{2}$: outside the range it overestimates, and resource 2 now has shadow price 0.
(d) $a = (1,1,1)^\top$: $B^{-1}a = (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})^\top$, $r_u = p - y^\top a = p - \frac{5}{2}$. Old basis optimal iff $p \le \frac{5}{2}$. For $p = 3$, $r_u = \frac{1}{2}$; rows: $x_2 = \frac{3}{2} - \frac{1}{2}u - \ldots$, $x_1 = \frac{5}{2} - \frac{1}{2}u - \ldots$, $s_3 = \frac{1}{2} + \frac{1}{2}u + \ldots$; ratios 3 and 5, so x_2 leaves at $u = 3$. New optimum $x_1 = \frac{5}{2} - \frac{3}{2} = 1$, $x_2 = 0$, $u = 3$, $s_3 = 2$, $z^\star = \frac{21}{2} + 3 \cdot \frac{1}{2} = 12$. Check with the multipliers of the new basis (u, x_1, s_3): $B^\top y = c_B$ gives $y = (3, 0, 0)$; reduced costs x_2: $2 - 3 = -1$, x_3: $4 - 6 = -2$, s_1: -3, s_2: 0; none positive, so 12 is optimal, and the zero on s_2 at this nondegenerate BFS means the optimum is not unique.
(e) Substitute $x_1 = \frac{5}{2} - \frac{3}{2}x_3 - \frac{1}{2}s_2$ into $z + \gamma x_1$:

$$z = \tfrac{21}{2} + \tfrac{5}{2}\gamma - \tfrac{3}{2}(1+\gamma)x_3 - 2s_1 - \tfrac{1}{2}(1+\gamma)s_2,$$

optimal iff $1 + \gamma \ge 0$: $\gamma \ge -1$, no upper end. Increasing the profit of a variable that is already basic can only make its reduced-cost tests easier (every x_1-coefficient in the row is negative, so $-\gamma\times$(negative) stays negative); the plan $(\frac{5}{2}, \frac{3}{2}, 0)$ remains optimal for every $\gamma \ge -1$, with value $\frac{21}{2} + \frac{5}{2}\gamma$. At $\gamma = -1$ two reduced costs vanish and the basis (x_2, x_3, s_3) of tableau 2 is also optimal, value 8.

Q6

1. False: $\max\{-x_1 : x_1 \ge 0\} = 0$. Bounded *values* is the question, not a bounded set.
2. False: at most m; a degenerate BFS has fewer (exercise 17d: $(0, 16, 0, 0)$ with $m = 2$).
3. True: the objective row $z = z_0 + \sum r_j x_j$ with $r \le 0$ and $x \ge 0$ bounds every feasible value by z_0.
4. False: at a degenerate optimum one basis can fail (exercise 16d); the converse needs nondegeneracy.
5. False: both can be infeasible (the pair with rows $x_1 - x_2 \le -1$, $-x_1 + x_2 \le -1$).
6. False: $y_i s_i = 0$ allows both to be zero; tightness does not force $y_i > 0$ (exercise 8, staff row).

7. True: $r_j = c_j - c_B^\top B^{-1} A_j$ does not involve b.
8. False: $z^\star = c_B(\gamma)^\top B^{-1} b(\delta)$ contains a $\gamma\delta$ term (exercise 17: $11 \neq 5 + 8$).
9. True: move along the zero-cost basic direction; nondegeneracy gives a positive feasible step with constant objective (exercise 16e).
10. False: the feasible set is unchanged, so the optimal value is unchanged; only the number of slack variables changes.